CALCULATION-04

CUBE (घन)

CLASS NOTES

CUBE / घन

$1^3 = 1$

$2^3 = 8$

$3^3 = 27$

$4^3 = 64$

$5^3 = 125$

$6^3 = 216$

$7^3 = 343$

$8^3 = 512$

$9^3 = 729$

$10^3 = 1000$

$11^3 = 1331$

$12^3 = 1728$

$13^3 = 2197$

$14^3 = 2744$

$15^3 = 3375$

$16^3 = 4096$

$17^3 = 4913$

$18^3 = 5832$

$19^3 = 6859$

$20^3 = 8000$

$21^3 = 9261$

$25^3 = 15625$

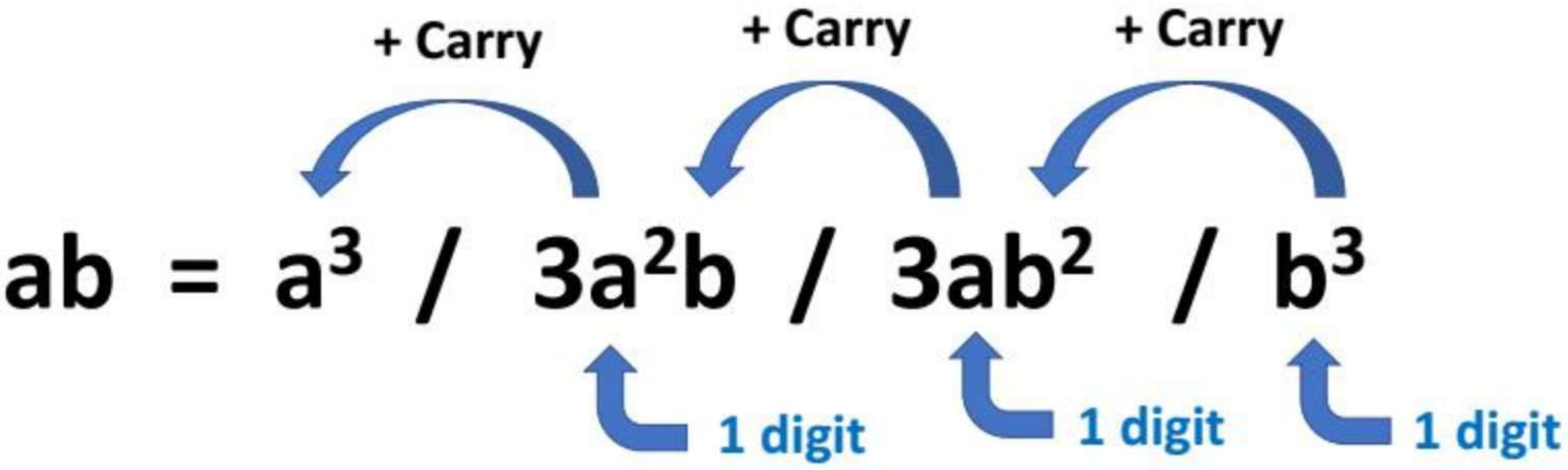
$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(23)^3 = 2^3 / 3 \times 2^2 \times 3 / 3 \times 2 \times 3^2 / 3^3$$

$$= 8 / 36 / 54 / 27$$

$$= 12167$$

CUBE OF A TWO DIGIT NUMBER
दो अंकों की संख्या का घन



Find the cube of 14

Here, a = 1 & b = 4

$14^3 = 1^3 / 3.1^2.4 / 3.1.4^2 / 4^3$

$= 1 / 12 / 48 / 64$

$= 2744$

Find the cube of 23

Here, a = 2 & b = 3

$23^3 = 2^3 / 3.2^2.3 / 3.2.3^2 / 3^3$

$= 8 / 36 / 54 / 27$

$= 12167$

Find the cube of 38

Here, a = 3 & b = 8

$38^3 = 3^3 / 3.3^2.8 / 3.3.8^2 / 8^3$

$= 27 / 216 / 576 / 512$

$= 54872$

Find the cube of 79

Here, a = 7 & b = 9

$79^3 = 7^3 / 3.7^2.9 / 3.7.9^2 / 9^3$

$= 343 / 1323 / 1701 / 729$

$= 493039$

Find the cube of 79

Here, a = 7 & b = 9

$79^3 = 7^3 / 3.7^2.9 / 3.7.9^2 / 9^3$

$= 343 / 1323 / 1701 / 729$

$= 493039$

16^3 = 1 / 18 / 108 / 816

= 4096

34^3 = 27 / 108 / 144 / 64

= 39304

84^3 = 512 / 768 / 384 / 64

592704

62^3 = 216 | 216 | 72 | 8

= 238328

71^3 = 343 | 147 | 21 | 1

357911

22^3 = Ans in Comment Box

87^3 =

69^3 =

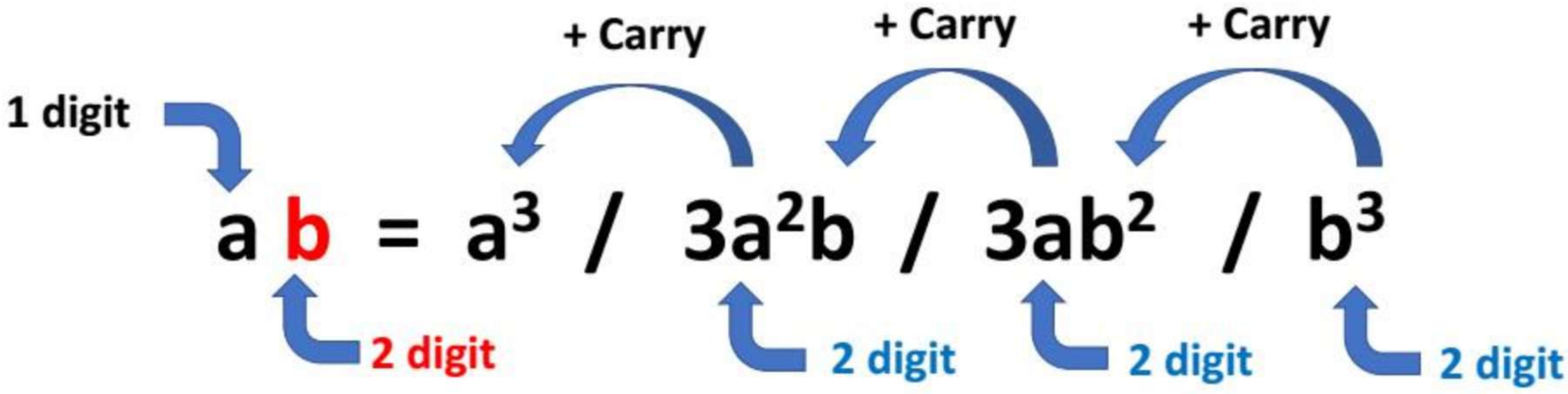
43^3 =

51^3 =

59^3 =

74^3 =

CUBE OF A THREE DIGIT NUMBER
तीन अंकों की संख्या का घन



$$(\underset{a}{1}\underset{b}{04})^3 = a^3 / 3a^2b / 3ab^2 / b^3$$

$$(104)^3 = 1 / 12 / 48 / 64$$

$$= 1\,12\,48\,64$$

Find the cube of 104

Here, a = 1 & b = 4

$104^3 = 1^3 / 3.1^2.4 / 3.1.4^2 / 4^3$

$= 1 / 12 / 48 / 64$

$= 1124864$

Find the cube of 206

Here, a = 2 & b = 6

$206^3 = 2^3 / 3.2^2.6 / 3.2.6^2 / 6^3$

$= 8 / 72 / 216 / 216$

$= 8741816$

Find the cube of 312

Here, a = 3 & b = 12

$206^3 = 3^3 / 3.3^2.12 / 3.3.12^2 / 12^3$

$= 27 / 324 / 1296 / 1728$

$= 30371328$

806^3 = 512/1152/864/216

= 523606616

506^3 = 125/450/540/216

= 129554216

109^3 =

$$408^3 = 4^3/3\times4^2\times8/3\times4\times8^2/8^3$$

$$= 64/384/768/512$$

$$= 64\ 91\ 73\ 12$$

$(509)^3 = 125/675/1215/729$

$131\ 872229$

$(904)^3 =$ Ans in Comment Box

$$\underbrace{(2}_{a}\underbrace{13)}_{b}^{3} = 2^3 / 3x2^2x13 / 3x2x(13)^2 / (13)^3$$

$$= 8 / 156 / 1014 / 2197$$

$$= \underline{9663597}$$

$(412)^3 = 64 / 576 / 1728 / 1728$

69 934 528

$$711^3 = 343|1617|2541|1331$$

$$359425431$$

$(912)^3$ = Ans in comment Box